

Approval gates and idempotency

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Planning and action selection

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Logs traces and privacy

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Integration tests and fixtures

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Approval gates and idempotency. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Approval gates and idempotency. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Key J already committed room A. A retry uses key J but room B. What should happen?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Key J already committed room A. A retry uses key J but room B. What should happen?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled approval gates and idempotency evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of approval gates and idempotency. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Approval gates and idempotency

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

Approval gates and idempotency

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check approval

The approved identifier and desk match K9.



2. Execute first copy

No result is stored yet, so perform the approved reservation once.



3. Handle duplicate

Return the stored K9 result without another reservation.

CHANGE ONE THING

A new K10 asks for desk 5 without approval.

Explain the changed result using the matching lesson.

Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return the stored K9 result without another reservation.

1. Execute first copy
2. Check approval
3. Handle duplicate

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Handle duplicate Return the stored K9 result without another reservation. Approval answers whether an action may proceed. Idempotency prevents repeated handling of that action from creating duplicate effects.

Practice 2: explain the deciding evidence

Changed-case practice: A new K10 asks for desk 5 without approval. Which conclusion is supported by the supplied facts?

1. Stop K10 at the approval gate.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Stop K10 at the approval gate. K9's approval does not apply to a different action.

Repair a missing building block

Review Planning and action selection (ai-6/14). Goal: Explain the mechanism and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Logs traces and privacy (ai-6/12). Goal: Explain the mechanism and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

Review Integration tests and fixtures (ai-6/9). Goal: Explain the mechanism and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
